

Vacuum popcorn: ultra-high-energy particles from collapsing false-vacuum pockets

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Abstract

First-order phase transitions in the early universe can leave behind pockets of false vacuum that trap a gas of particles. Inside a pocket the particles are light, and they lack the energy to take on their larger mass outside it. If such a pocket collapses today (a “pop”), its wall becomes ultra-relativistic and throws the trapped particles out, much as fingers pinched together shoot out a pip. I followed tens of thousands of them through such collapses with exact relativistic kinematics. A particle that meets the fast-moving wall with energy E leaves with about $m_{\text{out}}^2/4E$, where m_{out} is its mass outside. For a gas at rest and mass ratios up to a million, the median escapee carries 0.1 to 1.6 per cent of $m_{\text{out}}^2/m_{\text{in}}$. At larger ratios most escapees leave with less, but the most energetic tenth still carry 0.05 to 0.5 per cent of $m_{\text{out}}^2/m_{\text{in}}$. A particle of 10^7 GeV outside and 10 GeV inside has a median escape energy near 10^{20} eV, and one of 1 TeV outside and 10 eV inside sends a tenth or more of its escapees above that. If the escapees decay to photons, Auger’s photon limit caps pops in the Galaxy at about three events a year in a 3000 km^2 array. The nearest pops would arrive as several air showers at the same instant from one direction. No experiment has yet searched for such groups at these energies.

Keywords: Ultra-high-energy cosmic radiation, Cosmic ray sources, Cosmological phase transitions, Particle astrophysics, Dark matter, Cosmic ray showers

1 Ingredients

Air-shower arrays have recorded cosmic rays near 10^{20} eV since 1962 (Linsley 1963), and the sources of the most energetic ones are still unknown. Most proposed sources accelerate charged nuclei in shocks and jets; a few use relics of the early universe, such as collapsing cosmic textures (Brandenberger, Cyr & Jiao 2020) or vacuum bubbles seeded by merging black holes (Sakharov, Konoplich & Gogberashvili 2025). This letter’s recipe needs a pocket of false vacuum, a particle light inside the pocket and heavy outside it, and something that makes the pocket collapse today. Such a collapse is what this letter calls a pop.

Recent work supplies the pocket and the particle together. When a first-order phase transition gives a particle a large mass in the new vacuum, particles without the energy to take on that mass cannot enter it (Baker, Kopp & Long 2020). They pile up in the shrinking remnants of the old vacuum. Those remnants are the Fermi balls of Kawana & Xie (2022) and, where a hot axion gas holds the pocket open, the axion relic pockets of Carenza et al. (2024). What makes a pocket collapse today is less settled. Fermi balls can collapse late, into black holes, possibly as late as today (Lu, Kawana & Kusenko 2023; Picker & Kusenko 2023). Khater, Marsh & Nikolis (2026) note that the smallest axion relic pockets, which hold only a few axions, should classically collapse, and they leave that case for future work.

As far as I can find, no one has worked out what the trapped particles carry away when a pocket collapses. Lewicki et al. (2023), who simulated such collapses, found that the particles slow or even reverse them, and that a pocket can shrink to nothing only if they all escape first. This letter

computes what escapes, with what energy, and how Earth would see it: as air showers that arrive together (Figure 1).

2 Pinching a pip

Pinch a wet melon pip between finger and thumb and it flies off far faster than your fingers closed. A collapsing pocket does the same to the particles it traps, and its kinematics can be written down exactly.

Take a wall moving with Lorentz factor γ into a region where a particle has mass m_{in} , with mass $m_{\text{out}} > m_{\text{in}}$ behind the wall. In the wall’s frame energy is conserved, and the particle crosses only if its momentum normal to the wall, u , satisfies $u^2 > m_{\text{out}}^2 - m_{\text{in}}^2$. Otherwise the wall reflects it (Dine et al. 1992; Arnold 1993; Lewicki et al. 2023). A particle at rest that the wall reflects leaves with energy $\gamma^2(1 + \beta^2)m_{\text{in}} \approx 2\gamma^2 m_{\text{in}}$ (Kawana, Lu & Xie 2022; Garcia Garcia, Koszegi & Petrossian-Byrne 2023). Any wall conserves the particle’s momentum along it, $p_{\perp}$. A wall moving at nearly the speed of light also conserves its light-cone momentum $E + p_n$, where p_n is its momentum along the normal pointing out of the pocket (Bödeker & Moore 2009). Walls this fast have been studied as particle accelerators in the early universe (Baldes et al. 2025; Shakya 2024).

From the conservation of $E + p_n$, a particle at rest that such a wall sweeps up leaves with

$$E = \frac{m_{\text{out}}^2 + m_{\text{in}}^2}{2m_{\text{in}}}, \quad (1)$$

and one that meets the wall head-on with energy $E \gg m_{\text{in}}$

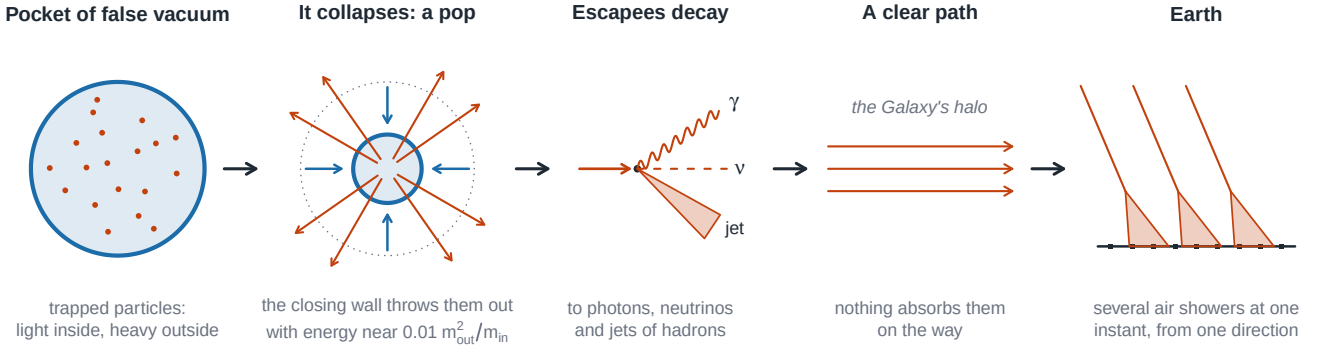


Figure 1: A pop, from pocket to air showers. A pocket of false vacuum holds particles that are light inside it and heavy outside. When it collapses, its wall throws them out. They decay, their products cross the Galaxy’s halo unabsorbed, and they reach Earth as air showers that arrive together from one direction.

leaves with

$$E' = E + \frac{m_{\text{out}}^2}{4E}. \quad (2)$$

The less energy a particle brings to the wall, the more it takes away; the wall’s motion supplies the difference. For any reflection, conserving energy in the wall’s frame ties the lab energies before and after it, now written E and E' , through $E E' = \gamma^2(m_{\text{in}}^2 + p_{\perp}^2) + u^2$, with u again the normal momentum in that frame. A particle reflected from rest therefore leaves with at most $(2m_{\text{out}}^2 - m_{\text{in}}^2)/m_{\text{in}}$, because a faster wall would let it through instead.

3 Following a collapse

A thin wall in flat space that starts from rest at radius R_0 has, by energy conservation, Lorentz factor

$$\gamma(x) = \frac{k(1-x^3)+1}{x^2}, \quad x = \frac{R}{R_0}, \quad k = \frac{\varepsilon R_0}{3\sigma}, \quad (3)$$

where ε is the vacuum energy released per unit volume and σ the wall tension. Small k means tension drives the collapse, large k means vacuum energy does. Near the centre γ grows as $1/R^2$, and a particle that the wall reaches late meets it moving close to the speed of light.

I followed trapped particles through such collapses one encounter at a time, letting each fly straight between encounters and applying the exact rule at each. A particle starts from a random point in the pocket, either at rest or with a thermal or degenerate momentum. One that starts at rest only ever moves along a radius, and the others I follow in three dimensions. A pocket ends when it has shrunk to its own wall thickness, a fraction x_{min} of its starting size; particles still inside then belong to the wall’s final burst, which this letter does not follow. Appendix A describes the numerical method and its checks.

4 What comes out

Figure 2 shows the escapees. For a gas at rest with $m_{\text{out}}/m_{\text{in}}$ between 10^4 and 10^6 , the median particle leaves with 0.5 to 1.6 per cent of $m_{\text{out}}^2/m_{\text{in}}$ when tension drives the collapse or shares it evenly ($k \leq 1$). It leaves with about 0.1 per cent when vacuum energy drives it ($k = 10$). The spread is one

to two decades between the 10th and 90th percentiles. Between half and nine-tenths of the escapees take the same route. A still-slow wall hits them once, giving them energy E_1 . They cross the centre and leave through the far side of the wall, by then ultra-fast, with between $E_1 + m_{\text{out}}^2/4E_1$ and $m_{\text{out}}^2/2E_1$, close to what equation (2) gives. Where each particle sat sets its E_1 , which is why the spread is wide. The rest bounce more than once, often after the wall catches them up from behind. When $m_{\text{out}}/m_{\text{in}} = 100$, up to 6 per cent of all escapees leave above the reflection limit of section 2, which holds only for head-on bounces.

At larger mass ratios the far side of the wall is often still too slow to let a particle through on its first return. The particle is thrown back again and gains energy at each extra bounce, and because equation (2) gives less to a particle that arrives with more, it finally leaves with less than a single bounce would have given it. The median then falls, by an amount that depends on how far the pocket shrinks. The most energetic tenth of the escapees still carry 0.2 to 0.5 per cent of $m_{\text{out}}^2/m_{\text{in}}$ at a ratio of 10^{10} and 0.05 to 0.2 per cent at 10^{12} . Those figures are for pockets that end between 10^{-12} and 10^{-15} of their starting size, the range that pockets of 10^{15} to 10^{20} g reach.

In a hot or degenerate gas the particles’ typical energy E_{typ} takes the place of m_{in} . The median escapee carries 18 to 23 per cent of $m_{\text{out}}^2/E_{\text{typ}}$ when $m_{\text{out}}/E_{\text{typ}} = 10^2$, and 1.9 to 2.9 per cent when it is 10^4 , for thermal and Fermi gases alike.

Because the outside mass enters squared, modest masses reach the highest energies. A particle that weighs 10^7 GeV outside a pocket and 10 GeV inside it, in a gas at rest with $k \leq 1$, has a median escape energy near 10^{20} eV, and about half its escapees exceed it. One of 1 TeV outside and 10 eV inside sends a tenth to a quarter of its escapees above 10^{20} eV. A hot gas with $m_{\text{out}}/E_{\text{typ}} = 10^4$ has a median near 10^{20} eV when $m_{\text{out}} \approx 4 \times 10^8$ GeV.

5 A pan of popcorn

What arrives at Earth depends on what the escapees turn into. Outside the pocket their extra mass opens decay channels that were closed inside, and only their decay products can be seen: photons, neutrinos and quark jets. The jets make mostly pions, which end as photons and neutrinos,

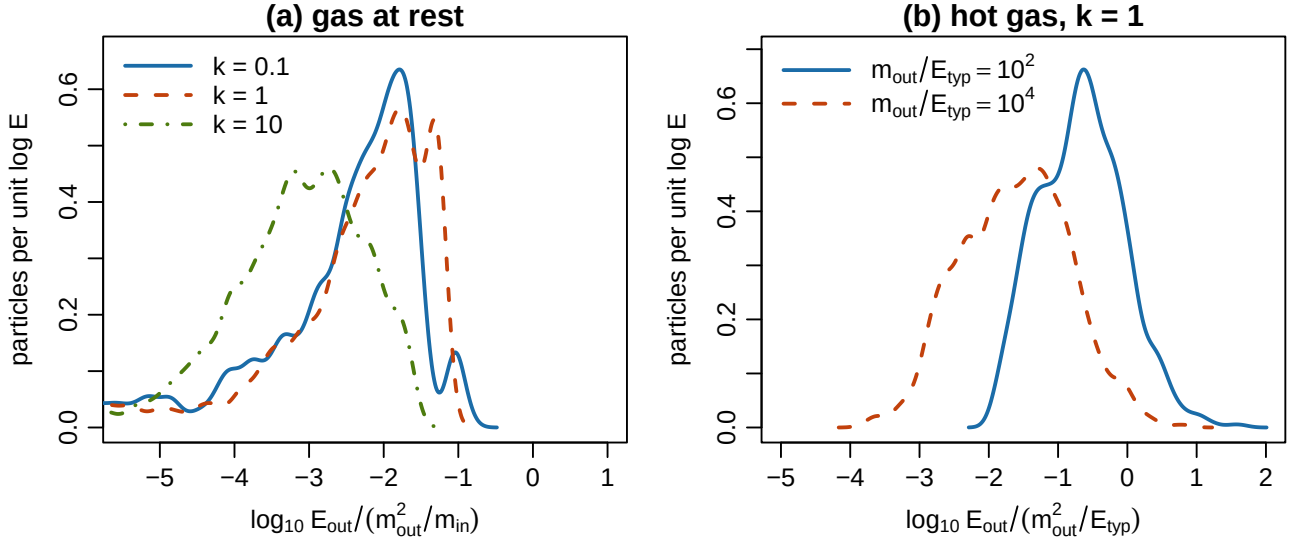


Figure 2: Energies of the particles that escape a collapsing pocket. (a) A gas initially at rest, $m_{\text{out}}/m_{\text{in}} = 10^6$, in a pocket that ends at 10^{-12} of its starting radius, for three wall histories. (b) A hot thermal gas, $k = 1$, for two ratios of m_{out} to the gas's typical energy.

with few nucleons and no heavier nuclei. Pops therefore cannot supply the bulk of the highest-energy cosmic rays, which Auger finds to be mostly heavier nuclei (Pierre Auger Collaboration 2017).

If the pockets are part of the Galaxy's dark matter, the Galaxy pops like a pan of popcorn, one pocket at a time, and photons from its pops reach us. Above 10^{18} eV their attenuation length exceeds the size of the halo (Kalashev & Kuznetsov 2016), and the shortest published interaction length at 10^{19} to 10^{20} eV, with the radio background included, is about 1.6 Mpc (Nițu et al. 2021). Suppose each pop puts a fraction f of its pocket's rest energy into escapees of 10^{20} eV that decay to photon pairs, at a rate Γ per pocket. Take a standard halo (Navarro, Frenk & White 1997; scale radius 20 kpc, 0.4 GeV cm^{-3} at the Sun). Auger's limit on photons above 40 EeV, $1.72 \times 10^{-4} \text{ km}^{-2} \text{ sr}^{-1} \text{ yr}^{-1}$ (Pierre Auger Collaboration 2023a), then caps $f\Gamma$ at $1.6 \times 10^{-32} \text{ s}^{-1}$, which allows no more than 7×10^{-15} of the pockets' rest energy to pop in a Hubble time. At that cap a 3000 km^2 array seeing the whole sky would record about 3 such events a year, whatever the pocket mass.

A few of the best-known events could have been pops. As a photon, only the 320 EeV Fly's Eye event of 1991 could (Risse et al. 2006). Telescope Array's 244 EeV event of 2021 was not a photon (Telescope Array Collaboration 2023), and nor were the highest-energy AGASA and Yakutsk events (Rubtsov et al. 2006). The jets' few nucleons include protons, each with about half to two photons at the same energy, by the fragmentation ratios of Aloisio, Berezhinsky & Kachelriess (2004). The searches that found no photons therefore allow two to ten of Telescope Array's 28 events above 100 EeV to be pop protons, and ten or more of Auger's 35 (Telescope Array Collaboration 2026; Pierre Auger Collaboration 2022, 2023b). The 244 EeV event, whose shower favoured a proton, can be one of them. Telescope Array's ten events of 2008 to 2013, whose directions are published (Telescope Array Collaboration 2014), also lean toward the Galactic centre, as halo pops would,

but Auger's 35 do not.

A pop also arrives all at once. Because its escapees leave within the collapse time and travel at nearly the speed of light, the decay products of one pop that reach Earth arrive together from one point. From 1 kpc away they arrive within 5 microseconds of each other if $m_{\text{out}} = 1 \text{ TeV}$, or within 5 seconds if $m_{\text{out}} = 10^6 \text{ GeV}$. At the photon cap, the share of events that arrive in simultaneous groups rises with the pocket mass. It is 0.4 per cent for pockets of 10^{15} g , 12 per cent for 10^{18} g and 70 per cent for 10^{20} g , about one group every two years in a 3000 km^2 array. I found no search by Auger or Telescope Array for simultaneous showers at these energies; Auger's multiplet searches look for steady sources lined up in arrival direction and inverse energy (Pierre Auger Collaboration 2012). The one search for coincident showers I know of, CHICOS above 10^{14} eV , reported "a 2.9σ excess observed for coincidence times less than $10 \mu\text{sec}$ " (Carlson et al. 2005). A particle of 100 GeV outside and 1 MeV inside would put a pop's showers near 10^{14} eV , in the range CHICOS searched, though nothing links that excess to pops.

6 What this leaves out

These results hold until the trapped particles have taken all the pocket's energy. For a gas at rest that comes when their number times $m_{\text{out}}^2/m_{\text{in}}$ is 10 to 80 times that energy in five of the six cases run, and more in the sixth. A heavier load stalls the wall (Lewicki et al. 2023). Reflection is treated classically, which needs a wall thicker than a particle's wavelength in the wall's frame. The thrown particle must also be stable inside the pocket and short-lived outside; that follows if its only Standard Model decays are to heavy states, which are closed inside, where the particle is light. What makes a relic pocket collapse today is still open.

The rate of pops is free, so no search can exclude them outright, but each claim can fail. An event shown to be heavier than a proton was not a pop. The 244 EeV event stops being a candidate if Telescope Array's photon search reaches

two to nine times its exposure above 10^{20} eV and finds no photon. And the lean can be tested now. Of Telescope Array’s 17 events above 100 EeV with no published direction (Telescope Array Collaboration 2023), an isotropic sky puts one within 50 degrees of the Galactic centre, and the lean of the published ones puts five.

Within those limits, a collapsing pocket throws out the particles it traps, the typical one with a per cent or less of $m_{\text{out}}^2/m_{\text{in}}$ and up to 0.2 per cent of them with more than $m_{\text{out}}^2/m_{\text{in}}$ itself, and TeV-scale masses can reach 10^{20} eV. A nearby pop would reach Earth as several showers within microseconds to seconds of each other, all from one point. Auger and Telescope Array record the time of every shower they detect, and their data could be searched for such groups.

Vacuum decay is usually pictured as the end of the universe, a bubble of lower vacuum that grows at the speed of light (Coleman & De Luccia 1980). A pop is that physics the other way round. The pocket holds the higher vacuum, so tension and vacuum energy both push its wall inward, away from us. This kind of vacuum decay ends the way popcorn does, with a pop, and all that arrives is a handful of air showers.

A Numerical method and checks

At $\gamma = 10^{14}$ a wall moves within five parts in 10^{29} of the speed of light, below the resolution of double-precision arithmetic, and the lag $1/\beta - 1$ cannot be found by subtracting β from 1. The tracker carries that lag, and each particle’s light-cone momentum, as quantities in their own right, and it reduces every timing question to integrals of the lag. Its encounter formulas agree with 100-digit arithmetic to 5×10^{-12} .

For particles at rest the tracker agrees with brute-force time stepping in 36 test collapses, to 3.3×10^{-7} . Leaving out the catch-ups, the encounters in which the wall overtakes a particle from behind, changes the result in 9 of 18 of those tests. The three-dimensional tracker reproduces the radial one to 2×10^{-10} , and its largest disagreement with brute force, 1.2×10^{-3} , falls to 1.7×10^{-5} when the brute-force time step is cut 25-fold.

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I used Claude Opus 5.5 (Anthropic, run in Claude Code, 30 September to 2 October 2026) to write and check the simulation code, to search the literature and to draft text. Claude Sonnet 5.5 subagents read the cited sources. MiniMax-M3 and DeepSeek (deepseek-reasoner), through their APIs, proofread each paragraph in repeated rounds, and I accepted or refused each note. Scripts checked every number against the code’s output and every reference against INSPIRE, and each numerical method was checked against an independent one, with errors planted to show the check could fail. I reviewed and revised everything and am responsible for it.

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Conflicts of Interest

None.

Data Availability

Every number in this letter is produced by a script named for the passage it supports; the code is at <https://github.com/BenWiseman/vacuum-popcorn>.

References

- Aloisio, R., Berezhinsky, V., & Kachelriess, M. 2004, Fragmentation functions in SUSY QCD and UHECR spectra produced in top - down models. *Phys. Rev. D*, 69, 094023. arXiv:hep-ph/0307279.
- Arnold, P. B. 1993, One loop fluctuation - dissipation formula for bubble wall velocity. *Phys. Rev. D*, 48, 1539. arXiv:hep-ph/9302258.
- Baker, M. J., Kopp, J., & Long, A. J. 2020, Filtered Dark Matter at a First Order Phase Transition. *Phys. Rev. Lett.*, 125, 151102. arXiv:1912.02830.
- Baldes, I., Dichtl, M., Gouttenoire, Y., & Sala, F. 2025, Ultrahigh-Energy Particle Collisions and Heavy Dark Matter at Phase Transitions. *Phys. Rev. Lett.*, 134, 061001. arXiv:2306.15555.
- Bödeker, D. & Moore, G. D. 2009, Can electroweak bubble walls run away? *JCAP*, 05, 009. arXiv:0903.4099.
- Brandenberger, R., Cyr, B., & Jiao, H. 2020, Cosmic Rays and Spectral Distortions from Collapsing Textures. *JCAP*, 09, 035. arXiv:2005.11099.
- Carenza, P., Eby, J., Iarygina, O., & Marsh, M. C. D. 2024, Axion relic pockets — a theory of dark matter. *JHEP*, 09, 023. arXiv:2407.01676.
- Carlson, B. E., Brobeck, E., Jillings, C. J., Larson, M. B., Lynn, T. W., & McKeown, R. D. 2005, Search for correlated high energy cosmic ray events with CHICOS. *J. Phys. G*, 31, 409. arXiv:astro-ph/0411212.
- Coleman, S. & De Luccia, F. 1980, *Phys. Rev. D*, 21, 3305.
- Dine, M., Leigh, R. G., Huot, P. Y., Linde, A. D., & Linde, D. A. 1992, Towards the theory of the electroweak phase transition. *Phys. Rev. D*, 46, 550. arXiv:hep-ph/9203203.
- Garcia Garcia, I., Koszegi, G., & Petrossian-Byrne, R. 2023, Reflections on bubble walls. *JHEP*, 09, 013. arXiv:2212.10572.
- Kalashev, O. K. & Kuznetsov, M. Y. 2016, Constraining heavy decaying dark matter with the high energy gamma-ray limits. *Phys. Rev. D*, 94, 063535. arXiv:1606.07354.

- Kawana, K., Lu, P., & Xie, K.-P. 2022, First-order phase transition and fate of false vacuum remnants. *JCAP*, 10, 030. arXiv:2206.09923.
- Kawana, K. & Xie, K.-P. 2022, Primordial black holes from a cosmic phase transition: The collapse of Fermi-balls. *Phys. Lett. B*, 824, 136791. arXiv:2106.00111.
- Khater, W., Marsh, M. C. D., & Nikolis, C. 2026, The phenomenology of Axion Relic Pockets. arXiv:2609.24714.
- Lewicki, M., Mürsepp, K., Pata, J., Vasar, M., Vaskonen, V., & Veermäe, H. 2023, Dynamics of false vacuum bubbles with trapped particles. *Phys. Rev. D*, 108, 036023. arXiv:2305.07702.
- Linsley, J. 1963, *Phys. Rev. Lett.*, 10, 146.
- Lu, P., Kawana, K., & Kusenko, A. 2023, Late-forming primordial black holes: Beyond the CMB era. *Phys. Rev. D*, 107, 103037. arXiv:2210.16462.
- Navarro, J. F., Frenk, C. S., & White, S. D. M. 1997, A Universal density profile from hierarchical clustering. *ApJ*, 490, 493. arXiv:astro-ph/9611107.
- Nițu, I. C., Bevins, H. T. J., Bray, J. D., & Scaife, A. M. M. 2021, An updated estimate of the cosmic radio background and implications for ultra-high-energy photon propagation. *Astropart. Phys.*, 126, 102532. arXiv:2004.13596.
- Picker, Z. S. C. & Kusenko, A. 2023, Constraints on late-forming exploding black holes. *Phys. Rev. D*, 108, 023012. arXiv:2305.13429.
- Pierre Auger Collaboration 2012, Search for signatures of magnetically-induced alignment in the arrival directions measured by the Pierre Auger Observatory. *Astropart. Phys.*, 35, 354. arXiv:1111.2472.
- Pierre Auger Collaboration 2017, Combined fit of spectrum and composition data as measured by the Pierre Auger Observatory. *JCAP*, 04, 038. arXiv:1612.07155.
- Pierre Auger Collaboration 2022, Arrival Directions of Cosmic Rays above 32 EeV from Phase One of the Pierre Auger Observatory. *ApJ*, 935, 170. arXiv:2206.13492.
- Pierre Auger Collaboration 2023a, Search for Photons above 10^{19} eV with the Surface Detector of the Pierre Auger Observatory. *JCAP*, 05, 021. arXiv:2209.05926.
- Pierre Auger Collaboration 2023b, A Catalog of the Highest-Energy Cosmic Rays Recorded during Phase I of Operation of the Pierre Auger Observatory. *ApJS*, 264, 50. arXiv:2211.16020.
- Risse, M., Homola, P., Engel, R., Gora, D., Heck, D., Pekala, J., Wilczynska, B., & Wilczynski, H. 2006, On the primary particle type of the most energetic Fly's Eye event. *Nucl. Phys. B Proc. Suppl.*, 151, 96. arXiv:astro-ph/0410739.
- Rubtsov, G. I., Dedenko, L. G., Fedorova, G. F., Fedunin, E. Y., Glushkov, A. V., Gorbunov, D. S., Makarov, I. T., Pravdin, M. I., Roganova, T. M., Sleptsov, I. E., & Troitsky, S. V. 2006, Upper limit on the ultrahigh-energy photon flux from agasa and yakutsk data. *Phys. Rev. D*, 73, 063009. arXiv:astro-ph/0601449.
- Sakharov, A. S., Konoplich, R., & Gogberashvili, M. 2025, Ultrahigh-energy neutrino event KM3-230213A as a signal of electroweak vacuum turbulence in merging black hole binaries. *Phys. Rev. D*, 112, 083061. arXiv:2506.23387.
- Shakya, B. 2024, Cosmic Colliders: High Energy Physics with First-Order Phase Transitions. arXiv:2412.18752.
- Telescope Array Collaboration 2014, Indications of Intermediate-Scale Anisotropy of Cosmic Rays with Energy Greater Than 57 EeV in the Northern Sky Measured with the Surface Detector of the Telescope Array Experiment. *ApJ*, 790, L21. arXiv:1404.5890.
- Telescope Array Collaboration 2023, An extremely energetic cosmic ray observed by a surface detector array. *Science*, 382, 903. arXiv:2311.14231.
- Telescope Array Collaboration 2026, Searching for EeV photons with Telescope Array surface detector and neural networks. *JCAP*, 06, 052. arXiv:2512.01638.